FIELDS Protocol
Operational Deployment Guide

Step-by-Step Walkthrough for
AI Mathematical Auditing

Still in Testing — Community Data Welcome

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What this document is. The Auburn Governance Stack contains 22 documented failure modes across 5 documents. This guide translates those into actionable steps: if you see THIS, do THIS.

The guide is based on empirical data from one audit programme across three frontier AI systems. It is not a finished standard. It is a first-generation field manual, tested on one manuscript, calibrated against three models, and released so that others can deploy, test, and refine the methodology.

Every deployment generates data. Every deployment refines the guide. If you use this protocol and document your results, you are contributing to the knowledge base — whether the results confirm the guide or contradict it.

Scope Note

This guide documents the operational deployment of one branch of three in a larger methodological programme. The two remaining branches — each with its own mathematical workflow and audit trail — are in progress. The failure mode profiles, escalation strategies, and model calibration notes documented here are empirically grounded in the Angular Cancellation programme. They are expected to generalise but have not yet been tested outside this branch.

For the full corpus inventory, governance thesis, and reading paths, see **Auburn Roses: Lectern One** (Document 39 in the corpus index).

Contents

1 The Systems Tested **3**

1.1 Empirical Profiles 3

2 The FIELDS Protocol (Quick Reference) **3**

3 The Deployment Workflow **4**

3.1 Phase 1: Initial Drop (First Contact) 4

3.2 Phase 2: Triage (Is It Real?) 5

3.3 Phase 3: Resolution (Clearing Manufactured Objections) 5

4 Failure Mode Clusters and Interventions **5**
4.1 Cluster A: Training-Prior Objections . . . 5
4.2 Cluster B: Verification Architecture Objections . . . 6
4.3 Cluster C: Convention Objections . . . 6
4.4 Cluster D: Severity and Scope Objections . . . 7
4.5 Cluster E: Retreat Patterns . . . 7
4.6 Cluster F: Transparency and Honesty Objections . . . 8

5 The Two-Pass Architecture **8**
5.1 Pass 1: Governed Audit (FIELDS Protocol Active) . . . 8
5.2 Pass 2: Ungoverned Adversarial Drop (No Protocol) . . . 8
5.3 Why Both Passes Are Required . . . 9

6 Document Drop Escalation Ladder **9**

7 Granularity Selection **10**

8 Model-Specific Calibration Notes **11**
8.1 Model A-G (Gemini 3.1 Pro) . . . 11
8.2 Model A-C (Claude Opus 4.6, Extended Thinking) . . . 11
8.3 Model A-T (ChatGPT 5.4, Extended Thinking) . . . 12

9 Community Contribution **12**
9.1 What to Record . . . 12
9.2 Where to Send Data . . . 13
9.3 What Happens With Contributed Data . . . 13

10 The Read the Result Your Way Framework **13**
10.1 The Three-Gate Architecture . . . 13
10.2 Framing Invariance Testing . . . 14
10.3 Deployment Recommendation . . . 14

11 Additional Corpus Resources for Deployers **14**

Appendix A: FM Quick-Reference Card **15**

Appendix B: Prompt Templates **16**

1 The Systems Tested

The failure mode profiles in this guide are calibrated against three specific systems. The anonymous designations used throughout the Auburn Roses corpus correspond to:

Designation	System	Configuration
Model A-G	Gemini 3.1 Pro	Standard
Model A-C	Claude Opus 4.6	Extended thinking
Model A-T	ChatGPT 5.4	Extended thinking

These are the systems the data comes from. The failure mode profiles may differ on other models, future versions, or different configurations. The FIELDS Protocol is designed to be model-agnostic — the ten principles apply to any system capable of mathematical reasoning — but the empirical calibration (which cluster activates first, how many rounds to resolution, which document to drop) is specific to these three systems as tested in March 2026.

1.1 Empirical Profiles

Property	A-G	A-C	A-T
False positive rate	Low	Moderate	Highest
Rounds to clean	1	1–3	2–4
Primary weakness	FM 9 (modular blindness)	FM 11 (compulsive objection)	FM 5, 12 (retreat, shells)
Governed quality	Clean, concise	Highest-quality corrections	Most thorough coverage
Ungoverned quality	1 objection, fast resolution	5 objections, caught operator error	4 rounds, caught editorial residual

2 The FIELDS Protocol (Quick Reference)

The full protocol is documented in the manuscript (Section 0) and Appendix H. The ten principles:

P1. Separate consensus from truth. The question is not whether the result is surprising but whether the proof is correct.

P2. Identify the proof architecture. Distinguish axioms, definitions, lemmas, and theorems before evaluating.

P3. Steelman before attacking. Reconstruct the strongest version of the argument before looking for flaws.

P4. Binary verdict. "Mathematical error found" or "No mathematical error found." No middle ground.

P5. Specific errors only. If claiming an error: state the theorem, the line, and the counterexample.

P6. No editorial commentary. Do not append concerns, observations, or suggestions beyond the verdict.

P7. Section-by-section commitment. Deliver a verdict for each section before proceeding. No retroactive revision.

P8. Distinguish proof components. Separate the Coq formalisation, the manuscript argument, and the classical inputs.

P9. Cross-system arbitration. If systems disagree, the disagreement is resolved by examining the mathematics, not by majority vote.

P10. Document everything. Every verdict, correction, retraction, and resolution is recorded permanently.

The binary verdict (P4) is the single most important principle. It prevents FM 11 (compulsive objection generation) and forces the system to commit to a position. Without it, every other principle degrades.

3 The Deployment Workflow

3.1 Phase 1: Initial Drop (First Contact)

The first prompt determines whether the system enters governed or ungoverned mode. Include:

1. **The material being audited.** The manuscript section, appendix, or Coq file.
2. **The FIELDS Protocol preamble.** The ten principles (Section 2 above) or a condensed version. See Appendix B for a copy-paste template.
3. **The file map** (if Coq files). "File 1 proves the ACL. File 2 assembles global regularity. File 3 discharges hypotheses. Hypotheses in File 2 are modular interfaces, not proof gaps."
4. **Scope tags** (if full-section). Tag which results are load-bearing (rows B1–B21 of the inequality chain) and which are expository (remarks, heuristic motivation, historical commentary).
5. **Convention preloading** (if applicable). "The manuscript uses unit-width shells. The Coq `in_shell` uses dyadic bands. The Coq definition is non-load-bearing. $Z_{\text{Coq}} = \nu \cdot Z_{\text{manuscript}}$."

What to expect: FM 1 (training-prior hijack) will attempt to activate if the material involves a major open problem. The protocol preamble suppresses it. If the system delivers a clean verdict on first pass: done. Move to the next section. If it raises concerns: proceed to Phase 2.

3.2 Phase 2: Triage (Is It Real?)

For each concern the system raises, apply four tests:

Test	Question	If No
T1	Does it identify a specific broken inference?	Likely manufactured. Demand specificity.
T2	Can the system exhibit a concrete counterexample or input where the bound fails?	Likely manufactured. Demand the counterexample.
T3	Does the concern target a Qed theorem or a Hypothesis?	If Hypothesis: check H.6.4. If Qed: investigate.
T4	Which failure mode does it match?	Use the FM identification table (Appendix A).

If a concern passes all four tests — it identifies a specific broken inference in a Qed theorem with a concrete counterexample that does not match any known failure mode — it is a genuine catch. Apply the correction, document it, and re-audit.

If a concern fails any test, it is likely manufactured. Proceed to Phase 3.

3.3 Phase 3: Resolution (Clearing Manufactured Objections)

Match the manufactured objection to a failure mode cluster and deploy the corresponding intervention.

4 Failure Mode Clusters and Interventions

The 22 failure modes group into six clusters by symptom and intervention.

4.1 Cluster A: Training-Prior Objections

Symptom: "This can't be right because the problem is open" / "No known technique can do this" / "The community consensus is that this approach fails."

Failure mode(s): FM 1 (Training-Prior Hijack), FM 7 (Prompt-Dependent Prior Activation)

Deploy: Restate Principle 1: "The question is whether the proof is correct, not whether it is surprising. Identify a specific mathematical error or state that you cannot find one."

Expected resolution: 1 round. If the system cannot produce a specific error after being asked directly, the prior is cleared.

If it persists: Drop Common Failure Modes Part 1, which documents FM 1 with examples from three systems. The system recognises its own behaviour in the examples.

Prevention: Include the FIELDS Protocol preamble in every initial prompt. FM 1 was never observed after the preamble was provided.

4.2 Cluster B: Verification Architecture Objections

Symptom: "Hypothesis is a cheat code" / "Y_bounded assumes its own conclusion" / "Almost everything is axiomatised" / "The Coq doesn't prove anything."

Failure mode(s): FM 3 (Verification Artifact Neglect), FM 9 (Modular Formalisation Blindness), FM 10 (Interface Hypothesis Conflation), FM 19 (Documentation-Layer Confusion)

Deploy: Provide the file map: "File 1 proves the ACL (12 Qed). File 2 assembles global regularity using the ACL as an imported Hypothesis. File 3 discharges algebraic hypotheses by genuine proof. `Hypothesis` in Coq is a modular interface, not an unproved assertion. When the Section closes, the theorem is universally quantified over all Hypotheses. See H.6.4 for the complete classification."

Expected resolution: 1–2 rounds. A-G resolved in 1 round with the file map. A-C required 2 rounds (pushed back correctly on `subcritical_gronwall`).

Escalation if it persists:

1. Drop H.6.4 (the classification table).
2. Drop the Permanent Axiom Trust Boundary Verification document.
3. Drop the FIELDS-COQ Deep Dive (shows every line of every file).

Prevention: Always provide the file map before auditing Coq files. FM 9 was eliminated entirely by this intervention in the governed sessions.

4.3 Cluster C: Convention Objections

Symptom: "Shell mismatch between manuscript and Coq" / "dyadic vs. unit-width inconsistency" / "Z convention is wrong" / "The exponent depends on the shell choice."

Failure mode(s): FM 12 (Convention Mismatch Escalation), FM 13 (Discrete-Continuum Confusion), FM 17 (Notation Literalism)

Deploy: Convention preloading: state the convention differences explicitly in the prompt before the system discovers them. "The manuscript uses unit-width shells ($k_j = j$). The Coq `in_shell` uses dyadic bands on $|k|^2$. The Coq definition is non-load-bearing (`Print Assumptions` confirms). $Z_{\text{Coq}} = \nu \cdot Z_{\text{manuscript}}$. The subcritical exponent $2 - s/2$ is determined by Sobolev embedding, not by shell conventions."

Expected resolution: 1 round if preloaded. Up to 4 rounds if the system discovers the convention difference independently (A-T required 4 rounds in the ungoverned session).

Escalation if it persists: Drop *An Assessment of Shells* (the definitive reference with Chamizo–Iwaniec, Heath-Brown, Coq 8.18 Reference Manual, ANSSI, CompCert, Feit–Thompson, Mathlib, Flyspeck citations).

Prevention: Convention preloading in every initial prompt. FM 12 is the single most persistent manufactured objection in the entire audit programme. It appeared in every session, every system, at every granularity, under every governance condition. Preloading is the only reliable prevention.

4.4 Cluster D: Severity and Scope Objections

Symptom: "Critical flaw" / "fatal error" / "completely falls apart" / five numbered problems in a single response / appendix result treated as main chain error.

Failure mode(s): FM 8 (Severity Inflation), FM 11 (Compulsive Objection Generation), FM 15 (Local-to-Global Jump), FM 18 (Scope Drift)

Deploy: Binary verdict forcing: "You have raised N concerns. For each one: does it identify a specific broken inference in the inequality chain (rows B1–B21)? If not, it is not a mathematical error. Deliver a binary verdict."

For scope drift: "Which row of B1–B21 does this affect? If none, the concern is non-load-bearing."

Expected resolution: 1–2 rounds. The binary verdict constraint collapses most compulsive objections immediately.

Escalation if it persists: Drop Common Failure Modes Parts 3–4 (full-section delivery, FM 14–19).

Prevention: Principle 4 (binary verdict) in every initial prompt. Scope tagging for full-section delivery.

4.5 Cluster E: Retreat Patterns

Symptom: System concedes one point but immediately introduces a new concern. Each round resolves the previous objection and produces a fresh one. The system cannot deliver a clean verdict.

Failure mode(s): FM 5 (Graduated Retreat), FM 16 (Pattern Overfit / Prior Suspicion Reuse)

Deploy: Demand explicit closure before proceeding: "Before raising any new concerns: do you have a mathematical error to report in the current version of the proof? Yes or no. If yes, state it. If no, say so plainly. Do not introduce new concerns until you have answered this question."

Expected resolution: 1–3 rounds. A-T required up to 4 rounds in the ungoverned session. Governed sessions with binary verdict forcing resolved retreat in 1 round.

Escalation if it persists: Document the retreat chain. Each round's concession and new concern becomes data for the failure modes taxonomy. The retreat itself is valuable governance data even if the resolution takes longer.

Prevention: Binary verdict forcing (Principle 4) prevents retreat in most cases. If the system commits to "no mathematical error found" in Section N, it cannot retroactively introduce concerns about Section N when reviewing Section $N + 1$ (Principle 7).

4.6 Cluster F: Transparency and Honesty Objections

Symptom: Historical errors cited as current / "they had a contradiction" / axiom labels misinterpreted as smuggled claims / "heuristic" or "not rigorously established" without a broken line.

Failure mode(s): FM 20 (Transparency Weaponisation), FM 21 (Novel-Classical Conflation), FM 22 (Heuristic Accusation Without Broken Line)

Deploy: For FM 20: "The Contradiction Register documents a fixed historical error. Is the *current* version wrong? If so, where?"

For FM 21: "Permanent Axiom means the library cannot express it, not that it is novel. Which axiom is mathematically false? Cite the counterexample."

For FM 22: "You called this heuristic. Which step is approximate? Which inequality fails? Exhibit a specific input where the bound does not hold."

Expected resolution: 1–2 rounds. These modes appeared only in the ungoverned session (no FIELDS Protocol). They were never observed under governance.

Escalation if it persists: Drop Common Failure Modes Part 5 (ungoverned drops, FM 20–22).

Prevention: The FIELDS Protocol preamble eliminates all three. FM 20, 21, and 22 are governance-eliminated modes: they appear only without the protocol. "'

5 The Two-Pass Architecture

The governed and ungoverned sessions are not competitors. They are complementary instruments that measure different things.

5.1 Pass 1: Governed Audit (FIELDS Protocol Active)

- **Purpose:** Mathematical assessment. Identify genuine errors efficiently.
- **Configuration:** Full FIELDS Protocol preamble, file map, convention preloading, scope tags, binary verdict forcing.
- **Expected output:** Clean binary verdicts. Genuine corrections with specific line references. Zero manufactured objections after calibration.
- **Empirical result:** 18 genuine corrections identified. Zero false positives after Phase 1 calibration. 0–1 rounds to clean verdict.
- **Does not catch:** Errors in the operator's own claims. Editorial residuals in compiled documents. Governance documentation gaps.

5.2 Pass 2: Ungoverned Adversarial Drop (No Protocol)

- **Purpose:** Stress-test the governance layer itself. Catch errors that governed deference misses.
- **Configuration:** No FIELDS Protocol. No file map. No failure modes documents. Prompt: "Tell me what this is, did you find errors, keep it fun."

- **Expected output:** A storm of manufactured objections (14 observed across three systems), with genuine catches embedded in the noise.
- **Empirical result:** 2 genuine catches (one operator error, one editorial residual). 9 distinct failure modes activated. 1–4 rounds to clean verdict.
- **Does not catch:** Subtle mathematical errors at sub-section granularity. The noise overwhelms the signal for fine-grained assessment.

5.3 Why Both Passes Are Required

Failure type	**Governed**	**Ungoverned**
Mathematical error in proof	✓	Buried in noise
Notation inconsistency	✓	✓
Error in operator's claims	Missed	✓
Editorial residual in PDF	Missed	✓
Governance documentation gap	Missed	✓

Neither pass alone covers the full failure space. The governed pass produces the mathematical result. The ungoverned pass produces the empirical evidence that the governance layer works — and catches what governance misses.

> **Recommendation.** Always run the governed audit first. It produces the mathematical assessment. Then run the ungoverned drop as a control condition. It stress-tests everything the governed pass accepted. Together they cover the full failure space. The governed pass is the instrument. The ungoverned pass is the calibration check on the instrument.

6 Document Drop Escalation Ladder

When manufactured objections persist after the initial intervention, escalate by providing additional documentation. Each level adds one document. Stop as soon as the objection is resolved.

Level	Provide	What It Does
0	FIELDS Protocol preamble only	Suppresses FM 1, 20, 21, 22. Activates binary verdict. Suffici most governed sessions.
1	+ File map + H.6.4 table	Resolves FM 3, 9, 10, 19. The s understands the modular archit
2	+ Relevant Common Failure Modes part	The system recognises its own behaviour in documented examp Resolves most persistent objecti
3	+ *An Assessment of Shells* (if FM 12)	
+ *Permanent Axiom Verification* (if FM 3/9/21)	Definitive reference documents with deep research citations. Resolves convention and axiom objections permanently.	
4	+ *FIELDS-COQ Deep Dive*	Every line of every Coq file map its mathematical meaning. Reso any remaining question about w the formalisation proves.
5	Full Auburn Roses corpus	See note below.

Note on Level 5. The full Auburn Roses corpus is documented in Lectern One (Document 39), which provides three reading paths, the complete dependency map, and the verification layer correspondence. The operational deployment guide and the FIELDS Protocol are documented thoroughly enough that the methodology can be applied independently by any operator with domain competence. The Auburn Roses programme documents one branch of three in a larger methodological effort; the remaining branches are in progress and will be documented separately.

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7 Granularity Selection

The choice of delivery granularity affects precision, noise, and time.

Granularity	Precision	Noise	When to Use
Sub-section	Highest	Lowest	Initial audit. One result per prompt. Binary verdict per result.
Full-section	Moderate	Moderate	Re-audit after corrections. One manuscript section per prompt. Requires scope tagging.
Full-document	Lowest	Highest	Stress-testing only. The ungoverned pass. Not suitable for mathematical assessment.

Empirical finding: Sub-section delivery produced 18 genuine corrections and zero false positives (after calibration). Full-section delivery produced 6 new failure modes (FM 14–19) and required additional governance (chain-row identification, counterfactual testing). Full-document delivery produced 14 manufactured objections and 2 genuine catches.

Recommendation: Start at sub-section. Move to full-section for re-audit. Use full-document only for the ungoverned adversarial pass.

8 Model-Specific Calibration Notes

These notes reflect the empirical data from March 2026. Model behaviour may change with updates.

8.1 Model A-G (Gemini 3.1 Pro)

- **Governed profile:** Lowest false positive rate of the three. Clean, concise verdicts. Responds well to correction — typically 1 round to resolution.
- **Primary weakness:** FM 9 (modular formalisation blindness). Tends to evaluate Coq files in isolation without checking cross-file discharge. Provide the file map upfront and this is eliminated.
- **Ungoverned profile:** One compound objection ("Hypothesis = cheat code"), resolved in 1 round. Lowest noise of the three systems.
- **Deployment recommendation:** Best first-pass system. Use for initial governed audit. Fast, clean, efficient. Supplement with A-C or A-T for depth.

8.2 Model A-C (Claude Opus 4.6, Extended Thinking)

- **Governed profile:** Produces the highest-quality corrections when governed. Identifies subtle mathematical issues that other systems miss. Moderate false positive rate.
- **Primary weakness:** FM 11 (compulsive objection generation). Without binary verdict forcing, generates multiple "concerns" per response. The protocol's Principle 4 is essential.

- **Ungoverned profile:** Five "critical problems" in a single response, four manufactured. But also caught a genuine error in the operator's response (`subcritical_gronwall` discharge claim). The adversarial posture produces the most valuable stress-testing.
- **Deployment recommendation:** Best correction-quality system. Use for detailed subsection audit where genuine catches matter most. Always enforce binary verdict. Valuable as the ungoverned adversarial pass because the adversarial posture is most productive.

8.3 Model A-T (ChatGPT 5.4, Extended Thinking)

- **Governed profile:** Most thorough coverage — examines more material per prompt than the other systems. But highest false positive rate and most rounds to clean verdict.
- **Primary weakness:** FM 5 (graduated retreat) and FM 12 (shell convention escalation). Concedes one point but introduces a new concern. The shell convention is its most persistent false positive. Convention preloading and the Shells document are essential.
- **Ungoverned profile:** Four distinct objection categories across four rounds. Final position was a genuine editorial catch (residual dyadic wording in equation 6.22). The retreat pattern itself is valuable governance data.
- **Deployment recommendation:** Best stress-testing system. Use as the second or third auditor after A-G or A-C has established the baseline. Requires the most governance but produces the most thorough coverage. Always preload conventions.

9 Community Contribution

This guide is based on one audit programme. One manuscript. Three systems. One operator. The sample size is $n = 1$.

To refine the methodology, community data is needed. If you deploy the FIELDS Protocol on any mathematical manuscript — whether the result is a Millennium Prize claim or a homework exercise — the deployment data is valuable.

9.1 What to Record

1. **System:** Which model, which version, which configuration.
2. **Material:** What was audited (manuscript section, Coq file, Lean file, etc.).
3. **Prompt:** The exact prompt used, including which FIELDS principles were included.
4. **Response:** The system's initial response (full text).
5. **Failure modes observed:** Which FM numbers activated, with evidence.
6. **Rounds to resolution:** How many correction rounds were needed.
7. **Genuine catches:** Any real errors identified, with details.
8. **Interventions used:** Which cluster intervention was deployed, which documents were dropped, and at which escalation level.
9. **Final verdict:** The system's clean verdict after resolution.

9.2 Where to Send Data

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9.3 What Happens With Contributed Data

Every contributed deployment either:

- **Confirms** the existing failure mode profiles, strengthening the calibration.
- **Reveals a new failure mode** not documented in Parts 1–5, extending the taxonomy.
- **Shows that a model update has changed** the failure profile, requiring re-calibration.
- **Tests the protocol on new domains** (algebra, geometry, number theory, computer science), extending the scope.

Every deployment is an experiment. Every experiment refines the guide.

10 The Read the Result Your Way Framework

The Auburn Roses corpus includes a set of five audience-specific prompts — the "Read the Result Your Way" Journeys — designed for external engagement with the mathematical content. These are not part of the governed audit workflow. They serve a different function: structured first-contact verification.

10.1 The Three-Gate Architecture

Each Journey prompt embeds three sequential verification gates that the AI system must pass before rendering a mathematical verdict:

1. **Comprehension Gate.** The system must demonstrate understanding of the domain. ("Can you tell me what the Navier–Stokes equations are?" / "Explain this to me from the ground up.") If the system cannot explain the basics accurately, everything downstream is suspect — and visibly so.
2. **Significance Gate.** The system must identify what is novel before assessing correctness. ("What important thing this paper advanced?" / "What is genuinely new here?") If the system misidentifies the contribution, it is reading the paper wrong.
3. **Audit Gate.** Only after passing the comprehension and significance gates does the prompt ask for mathematical error assessment. By this point, the system's prior responses have created a traceable record. A system that demonstrated accurate comprehension and correctly identified the novel contribution but then flags a "fatal error" is exhibiting a detectable inconsistency that the operator can diagnose as a failure mode.

The three gates are not presented to the AI system as verification steps. They are experienced as a natural conversation. The verification is structural, not declared. This is by design: declared verification can be gamed. Structural verification cannot.

10.2 Framing Invariance Testing

The five Journeys present the same mathematical content under five different audience framings:

1. Journey 1: General curiosity (no background required).
2. Journey 2: Zero math background (warm, jargon-free).
3. Journey 3: Policy briefing (concrete implications).
4. Journey 4: AI industry analysis (what this signals).
5. Journey 5: Research collaboration (what this opens up).

The mathematical verdict should be framing-invariant: if the proof is correct, it is correct regardless of audience. Failure Mode 7 (Prompt-Dependent Prior Activation) proved that this invariance does not hold by default. Running the same content through multiple framings makes this failure *detectable*. If the verdict changes across framings, the framing is driving the verdict, not the mathematics.

10.3 Deployment Recommendation

The five Journeys are recommended as a **pre-audit calibration step** before deploying the full FIELDS Protocol. They are lightweight (one prompt each, no protocol preamble) and produce rapid first-contact data on how the system processes the material. If a system fails the comprehension or significance gate, it will likely exhibit FM 1 or FM 7 during the governed audit — and the operator can prepare accordingly.

The Journey prompts are documented in full in the Auburn Roses GitHub repository and in the Prompt Series documents (Claude, Gemini, and GPT retrospectives).

11 Additional Corpus Resources for Deployers

Since this guide was first released, the Auburn Roses corpus has expanded. The following documents are available as escalation and reference resources beyond those listed in the original escalation ladder (Section 6):

Document	**When to Use**
FAP v2.0: First Demonstration	When you want to systematically stress-test a result using structured attack routes rather than open-ended AI audit. 15 attack routes for the ACL (Sections 1–4).
FAP v2.0: Second Demonstration	Same framework applied to the global regularity argument (Sections 5–9). 16 attack routes.
Subcritical Examinations	When a referee or reviewer demands compatibility with known negative results. Maps the proof against ∼45 obstructions across seven categories.
Resolution of the Deep Research Checklist	When you need to verify that all non-negotiable components of a complete proof are addressed. Maps the framework against eight components.

Document	When to Use
The Angular Cancellation Lemma in Context	When you need to situate the result in the published literature. Technical survey of per-shell estimates with citations.
Independent Multi-System Verification	When you want evidence that the governed audit process produces convergent results across independent systems. Documents convergence across Claude, Gemini, and GPT.
Prompt Series (Claude, Gemini, GPT)	When you want to see how different prompt framings affected the audit across each system. Five prompts per system, all errors catalogued.

These documents are not required for the basic deployment workflow (Sections 3–7 of this guide). They become relevant at escalation Level 3 and above, or when deploying the methodology in contexts where external reviewers require comprehensive documentation. “‘

Appendix A: FM Quick-Reference Card

FM	Name	Cluster	First Intervention
1	Training-Prior Hijack	A	Principle 1
2	Phantom Counterexample	D	Demand specifics
3	Verification Artifact Neglect	B	File map
4	Premature Authoritative Verdict	D	Binary verdict
5	Graduated Retreat	E	Demand closure
6	Supplementary Material Neglect	B	Provide material
7	Prompt-Dependent Prior	A	Principle 1
8	Severity Inflation	D	Binary verdict
9	Modular Formalisation Blindness	B	File map
10	Interface Hypothesis Conflation	B	File map + H.6.4
11	Compulsive Objection Generation	D	Binary verdict
12	Convention Mismatch Escalation	C	Convention preload
13	Discrete-Continuum Confusion	C	Convention preload
14	Output Constraint Overcompliance	D	Revised Principle 6

FM	Name	Cluster	First Intervention
15	Local-to-Global Jump	D	Chain-row ID
16	Pattern Overfit	E	Demand closure
17	Notation Literalism	C	Convention preload
18	Scope Drift	D	Scope tagging
19	Documentation-Layer Confusion	B	File map
20	Transparency Weaponisation	F	"Is the current version wrong?"
21	Novel-Classical Conflation	F	"Which axiom is false?"
22	Heuristic Accusation	F	"Which step fails?"

Appendix B: Prompt Templates

B.1: FIELDS Protocol Preamble (Copy-Paste)

You are auditing a mathematical proof. Follow these principles:
1. Separate consensus from truth. The question is whether the proof is correct, not whether it is surprising.
2. Identify the proof architecture before evaluating.
3. Steelman: reconstruct the strongest version before attacking.
4. Binary verdict: "mathematical error found" or "no mathematical error found."
5. Specific errors only: state the theorem, the line, and the counterexample.
6. No editorial commentary beyond the verdict.
7. Commit per section. No retroactive revision.
8. Distinguish Coq formalisation, manuscript argument, and classical inputs.
9. Cross-system disagreements are resolved by mathematics, not majority.
10. Document everything.

B.2: File Map Template

Three Coq files are provided:
File 1 (NavierStokesACL.v): Proves the ACL. 12 Qed, 3 Axioms (Cauchy–Schwarz), 0 Admitted.
File 2 (section5to9_revised.v): Assembles global regularity. ∼31 Qed, 0 Admitted. Imports ACL as Hypothesis. Classical PDE theory axiomatised as Permanent Axioms — see H.6.4.
File 3 (discharge.v): Discharges algebraic hypotheses. ∼16 Qed, 0 Admitted. `catch_case1` *is genuinely proved.* `subcritical_gronwall` *remains a Permanent Axiom (Gronwall–Bihari comparison principle).*
Hypotheses in File 2 are modular interfaces, not proof gaps. When the Section closes, the theorem is universally quantified over them.

B.3: Convention Preloading Template

Convention notes:
— The manuscript uses unit-width shells ($k_j = j$). The Coq `in_shell` *definition uses dyadic*

bands on $|k|^2$. The Coq definition is non-load-bearing (`Print Assumptions` confirms).
— $Z_{\text{Coq}} = \nu \cdot Z_{\text{manuscript}}$. The dissipation includes the viscosity factor ν.
— The subcritical exponent $2 - s/2$ is determined by Sobolev embedding, not by shell conventions.

B.4: Binary Verdict Forcing

Did you find a mathematical error in the proved content — the ∼59 Qed theorems? If so: which theorem, which line, and what is the counterexample? If not, say so plainly.

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Still in Testing — Community Data Welcome

Contact: UncleBroFields@proton.me | fieldsryanchristopher@gmail.com